

RESEARCH ARTICLE

Advanced Agricultural Query Resolution Using Ensemble-Based Large Language Models

CYRENEO DOFITAS JR.¹, YONG-WOON KIM², AND YUNG-CHEOL BYUN³¹Department of Electronic Engineering, Institute of Information Science and Technology, Jeju National University, Jeju-si 63243, South Korea²Department of Computer Engineering, Jeju National University, Jeju-si 63243, South Korea³Department of Computer Engineering, Major of Electronic Engineering, Institute of Information Science and Technology, Jeju National University, Jeju-si 63243, South Korea

Corresponding author: Yung-Cheol Byun (ycb@jejunu.ac.kr)

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ABSTRACT Effective knowledge retrieval is crucial for addressing challenges related to optimization, such as pest management, soil health and crop productivity. Current single-model approaches struggle with limited accuracy, inconsistent responses, and inability to handle the increasing complexity of agricultural data, leading to unreliable recommendations for farmers. This study demonstrates an innovative weighted voting ensemble method that improves agricultural knowledge retrieval by combining Meta-LLaMA 3.1, Agricultural-BERT, and BERT-based-uncased. The ensemble model optimizes the prediction process by utilizing domain-specific data and a weighted voting mechanism to improve query performance and answer production. Our study outperformed individual models in providing accurate and contextually relevant responses, with an accuracy of 93%. We evaluated the system using both BLEU and ROUGE metrics to assess the quality of the generated text. Our ensemble model achieved a BLEU score 53.8 and demonstrated superior performance in the ROUGE evaluation, with ROUGE-1, ROUGE-2, and ROUGE-L scores of 0.70, 0.55 and 0.61. These results highlight the method's ability to generate contextually suitable, domain-specific responses that address the practical needs of agricultural experts. By integrating advanced LLMs with domain-specific knowledge, the proposed ensemble system significantly improves agricultural knowledge retrieval and provides more accurate and practical responses in the field. The findings suggest that the ensemble approach can effectively support decision-making in agricultural practices, particularly in management agricultural optimization.

INDEX TERMS Agricultural domain, BERT, Llama 3.1, agricultural-BERT, large language model, knowledge retrieval, weight voting average ensemble.

I. INTRODUCTION

Agriculture faces significant challenges in optimizing productivity, particularly in developing countries, where erratic weather, unreliable infrastructure, and pest invasions significantly reduce crop yields. Traditional knowledge retrieval systems often fail to manage the increasing complexity of agricultural data and struggle to provide accurate, actionable, and domain-specific insights. These limitations require innovative approaches to improve knowledge retrieval and decision-making within the agricultural sector.

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Large Language Models (LLMs) are designed algorithms capable of learning and generating human language. In agriculture, LLMs can analyze various data sources, such as soil reports, pest control procedures, and crop management strategies, to generate contextually relevant and actionable information [1], [2], [3]. Their ability to process extensive datasets and respond to complex queries sets them as transformative tools for agricultural knowledge retrieval [4], [5]. However, single-model LLMs often face limitations in handling domain-specific complexities and achieving consistent performance across diverse queries.

Ensemble techniques offer a promising solution by integrating multiple machine learning models to improve

predictive performance through complementary strengths [6]. These methods are particularly advantageous in the agricultural domain, where the integration of natural language processing (NLP) and LLM with emerging technologies can address challenges such as unpredictable weather, inadequate infrastructure, and pest invasions. By leveraging ensemble techniques, farmers can gain better insights to mitigate these challenges and enhance agricultural efficiency [7], [8]. Our study introduces a weighted voting ensemble approach that combines general-purpose LLMs, such as BERT-based uncased, with agriculture-specific models, such as Agricultural-BERT, and Meta-LLaMA 3.1. By integrating these diverse models, the ensemble system addresses the limitations of single-model approaches, enabling accurate and contextually relevant responses to complex agricultural queries.

This study demonstrates how the proposed ensemble approach can overcome these challenges, offering scalable, accurate, and context-specific solutions for agricultural knowledge retrieval. Using various LLM capabilities, the system improves decision-making, productivity, and sustainability in the agricultural sector [9]. In traditional approaches, the proposed system balances the complexity, relevance, and precision of the model, addressing the practical challenges faced by farmers while paving the way for improvements in agricultural optimization.

The LLMs in agriculture face challenges with domain-specific training data and high computing expenses, this study demonstrates how ensemble approaches can overcome these limitations. By offering accurate and scalable solutions for the retrieval of agricultural knowledge, this study improves agricultural decision-making, productivity, and sustainability [9]. Models such as BERT and LLaMA are versatile computational tools by being able to comprehend & act on natural language instructions. When the prompts are well designed, they can perform decently. Still, this approach introduces more complexity and hyperparameters to fine-tune, which makes it challenging to optimize and produces less consistent results. LLMs have generally transformed the landscape of text generation tasks, from machine translation to automatic summarization and more recently, data-to-text (D2T) generation. This model aims to convert semi-structured information, such as tables and graphs, into natural language text [10]. By leveraging a weighted voting ensemble, our study provides a scalable and innovative solution for addressing domain-specific challenges in agricultural knowledge retrieval. The approach integrates the strengths of general-purpose and domain-specific LLMs, enabling precise, contextually relevant responses for complex agricultural queries.

In healthcare domain, the prediction was carefully pre-processed, and four independent raters manually assessed the prediction of Bert model. This earlier study revealed solid inter-rater reliability and high performance in terms of grammatical accuracy and contextual appropriateness,

highlighting the potential of Bert model to support communication for people with language disorder [11].

Although LLMs offer unique capabilities, the effective deployment of such systems in agriculture presents several challenges, including the scarcity of domain-specific training data, high computational costs, and limited transferability between languages [12]. Advanced NLP techniques, such as Chain of Thought prompting and demonstration selection, have been used to fine-tune LLMs, improving their ability to extract relations and respond to queries [13]. However, further refinement is necessary to overcome these limitations and fully harness the potential of LLMs in agricultural applications. Our proposed weighted voting ensemble not only addresses the limitations of single-model techniques but also sets the stage for scalable, accurate, and context-specific solutions in agricultural knowledge retrieval. By combining diverse LLM capabilities, this system improves decision-making, productivity, and sustainability in the agricultural sector.

This study develops an ensemble of large-language models to improve agricultural knowledge retrieval and query resolution. Following are the major contributions of the suggested model:

- To create an LLMs and ensemble learning intuitive and farmer-centric interface that empowers farmers with actionable insights, predictive tools, and simplified access to user support.
- The study aims to improve agricultural yields and provide farmers with a practical system that will help them detect agricultural diseases and provide answers to queries related to crop production, thereby contributing to the growth of the agricultural sector.
- We utilize fine-tune ensemble LLM models to analyze textual data, identify crop symptoms, and provide accurate diagnoses, thereby reducing misdiagnosis and ensuring appropriate recommendations.

The remaining article is organized as follows. Section I discusses the study related to agriculture Section II discusses previous related work for agricultural knowledge. Section III presents material and methods including the proposed methodology, dataset acquisition, pre-processing techniques, and proposed model framework. Section IV presents the results and the proposed model comparison with prior related work and Section V we deeply discuss the approach using in the study and compared with existing study. Section VI concludes the research study.

II. RELATED WORK

Developing a reliable knowledge retrieval method involves addressing challenges such as improving LLMs' question-answering capabilities for effective source evaluation, extracting relevant content from large datasets and providing method stability despite information changes. This section summarizes the current state of LLM development and

the solutions offered by existing research to tackle these challenges.

Shows current models and methods for retrieving agricultural knowledge while highlighting their primary limitations. These include limited domain-specific accuracy, difficulty responding in real-time for practical applications, reliance on pre-trained models, and challenges in addressing complex agricultural environments that require fine-grained analysis. While AgriBERT and GPT-4 confront scalability and real-world application issues, models such as BERT-based-uncased and RoBERTa struggle with managing complex ontology structures or restricted emphasis areas. These limitations underscore the need for a more flexible and domain-specific solution, which lead to the development of our ensemble model, shown in Table 1.

GPT-4 was used for agricultural applications. However, it relied heavily on preexisting literature, limiting its ability to generate novel insights. Our approach overcomes this by incorporating domain-specific knowledge and real-time query processing. Recent work has shown the potential of LLMs in crop disease diagnosis, but their technique was limited to a single model [22], [23]. Our ensemble method aims to improve on this by leveraging the strengths of multiple models.

Recent research has shown diverse applications of LLMs in agriculture. ChatGPT and similar models have improved crop disease diagnosis, extension work, and plant health management decision support. However, these integrations face challenges in maintaining contextually appropriate outputs across different agricultural landscapes [24]. Knowledge graphs have been increasingly used to integrate genetic and omics data, delivering a comprehensive platform for translational research. For instance, the OrthoLegKB framework demonstrates how knowledge graphs integrate multispecies genetic data, allowing researchers to identify orthologs and candidate genes in legume species [25]. In addressing these challenges, transformer-based models such as GPT, BERT, and XLNet showed exceptional accuracy, exceeding conventional RNN and LSTM architectures that struggle with vanishing gradient issues. These algorithms are very successful in extracting domain-specific agricultural information [26]. Recent studies show that ensemble learning techniques that combining multiple pre-trained LLMs achieve higher accuracy in detecting and categorizing AI-generated content compared to single model approaches [27].

These large language models have also been used to improve agricultural meteorological suggestions. By employing multi-round prompt frameworks, these models iteratively improve suggestions based on feedback and new data, achieving high accuracy in providing farmers with actionable insights [28]. Additionally, this model has been used to improve agricultural weather forecasts. These models may iteratively enhance suggestions based on existing data and feedback by utilizing multi-round prompt frameworks, attaining high accuracy in providing farmers with helpful information [29].

LLMs have increased the quality and accuracy of information provided in various jobs in an innovative way. A new generative retrieval framework addresses dependability issues in LLMs when handling information retrieval tasks. This framework incorporates a generator, validator, and optimizer to ensure reliable and pertinent data sources capable of providing strong evidence for model outputs. Significant improvements in source verification and query expansion have been shown when these approaches are integrated, particularly with. Several studies have highlighted the need to advance agricultural search systems beyond basic data retrieval by incorporating result validation and optimization mechanisms [30], [31], [32]. To reduce the hallucinatory effects that frequently occur in LLMs, they utilize evidence-based validation procedures and feedback mechanisms. Compared to the information acquired from other domains, these methodologies have allowed LLMs to produce more accurate and relevant findings.

In the framework, they use a generator, validator, and optimizer. The generator obtains the relevant resources for the case and then evaluates its reliability and ability to back its conclusions with credible evidence. The optimizer then determines how valuable the information will be for your company. Using these methods has been demonstrated to outperform systems that concentrate on essential or difficult search jobs in terms of source validation and query extension. A previous study has examined the potential to use ensemble LLMs in several domains. For instance, studies show that combining the forecasts of multiple LLMs in real-world forecasting competitions leads to significantly better accuracy, rivaling human predictions [33]. LLM ensembles measure a decrease in single model bias and a gain in dependability, which is comparable to the increases in accuracy that humans crowd sourced.

LLMs have been used in agriculture to help farmers predict weather, diagnose pests and manage crops. Despite this, much research has yet to be done on ensemble approaches. This project aims to bridge the gap between unstructured data in agriculture and the precise insights that a farmer needs to take action by integrating LLM outputs. LLM ensembles can help address the growing demand for more intelligent, data-driven agricultural operations by offering more precise, reliable, and context-specific answers. Since then, pre-trained models (PTMs) have been extensively investigated, particularly for NLP and other data-intensive applications. An empirical investigation on model reuse within the HuggingFace community that highlights the main obstacles and advantages of PTM adoption [34]. They discovered that while PTM reuse may significantly accelerate the creation of AI apps, users need help understanding how models function, interpret the results they provide, and customize them for particular use cases. The difficulties with model selection and training pipeline concerns are among the most significant limitations, especially for novices. Compared to others, several publications have outlined the current situation

TABLE 1. Prior work on agriculture large language model.

Reffrences	Year	Models	Limitations
[14]	2021	Bert-based-uncased	Restrained by its emphasis on a narrow dataset of Amazon electronics reviews, reliance on binary sentiment classification, and the need for detailed error analysis or comparison with other models.
[15]	2022	Residual LSTM network and Sequence-to-Sequence	Low accuracy of existing agricultural Q&A model. Difficulty in meeting real-time user answer requirements.
[16]	2022	Bert	Investigates using food text for pre-training transformer-based models in agriculture.
[17]	2023	GPT-4 and Yolo	Inappropriate diagnosis and flawed solutions due to misidentification or misjudgment of agricultural concepts.
[18]	2024	RoBERTa	When domains or ranges are expressed as unions or intricate logical combinations of classes, higher ontology expressivity reflected in OWL may cause issues with this implementation.
[19]	2024	GPT-4	The primary drawback of this analysis is its dependence on an overview of intelligent farming technology from already published works and initiatives.
[20]	2024	AgriBERT	Data augmentation provides acceptable benefits for large pre-trained language models since the type these methods seek to deliver is frequently captured by their lengthy training.
[21]	2024	BERT-DPCNN	It performs exceptionally well when used with more diversified datasets complicated questions, or situations with limited resources.

regarding PTM reuse in other areas and have highlighted the need for more work to overcome technical obstacles and gain the support of the community [35], [36]. This method integrates statistical shrinkage techniques, such as LASSO and Elastic Net, to select and reduce base regressions, focusing on the most impactful models. Ensemble techniques like Multi-Layer Perceptron and Weighted Average are then used to aggregate the predictions, achieving higher predictive accuracy and lower computational costs [37]. A previous study introduces an ensemble-based model for sentiment analysis of persian comments on Instagram, leveraging deep learning algorithms to improve accuracy [38]. By integrating multiple models, the approach effectively captures the nuances of the Persian language, demonstrating the potential of ensemble techniques for language-specific challenges. This aligns with our study’s focus on using ensemble methods to improve domain-specific tasks, such as agricultural knowledge retrieval.

In other fields, other research has emphasized the importance of large language models (LLMs) in conjunction with ensemble learning techniques. This study has addressed diagnostic complexity and dataset limitations by creating weighted ensemble models for automated detection of diabetic retinopathy [39]. Similarly, an LLM-ensemble approach was introduced in the e-commerce industry to maximize the value extraction of product attributes. This method used the Dawid-Skene model to iteratively understand and assign weights to various LLM outputs, outperforming individual models in tasks like extracting attributes related to age and gender [40].

In addition to the e-commerce domain, ensemble learning has shown promise in other fields where LLMs are applied to complex data sets. Research in feature extraction using LLMs. In addition, it has further demonstrated the effectiveness of combining outputs from multiple LLMs to improve the precision of the extracted features [41], [42]. An especially important use-case is that of the agricultural knowledge retrieval, where the heterogeneity and complexity of queries demand a more complex technique than single-model approaches. Previous work has highlighted the need for domain boiling down any generic LLMs. Introduced BLADE framework: general LLMs are augmented with tiny domain-specific models, which is necessary for specialised tasks like in legal or medical tasks. Their work showed that by pretraining and fine-tuning smaller domain-specific models, BLADE is much better than traditional methods like retrieval augmentation [43].

LLMs provide replies that are more like those of actual humans, so they have been employed extensively in several industries. The majority, nevertheless, only dream and are unable to incorporate new information. Retrieval-augmented LLMs (such as those using the SURE framework) might help summarize pertinent external passages and produce more correct responses for ODQA tests to overcome these problems [44]. A machine learning technique called ensemble learning has been proposed, and research has demonstrated that utilizing it can also improve performance of the model [45]. These weights are further optimized by differential evolution to ensure that the ensemble is accurate and broadly applicable across various datasets. This technique has been

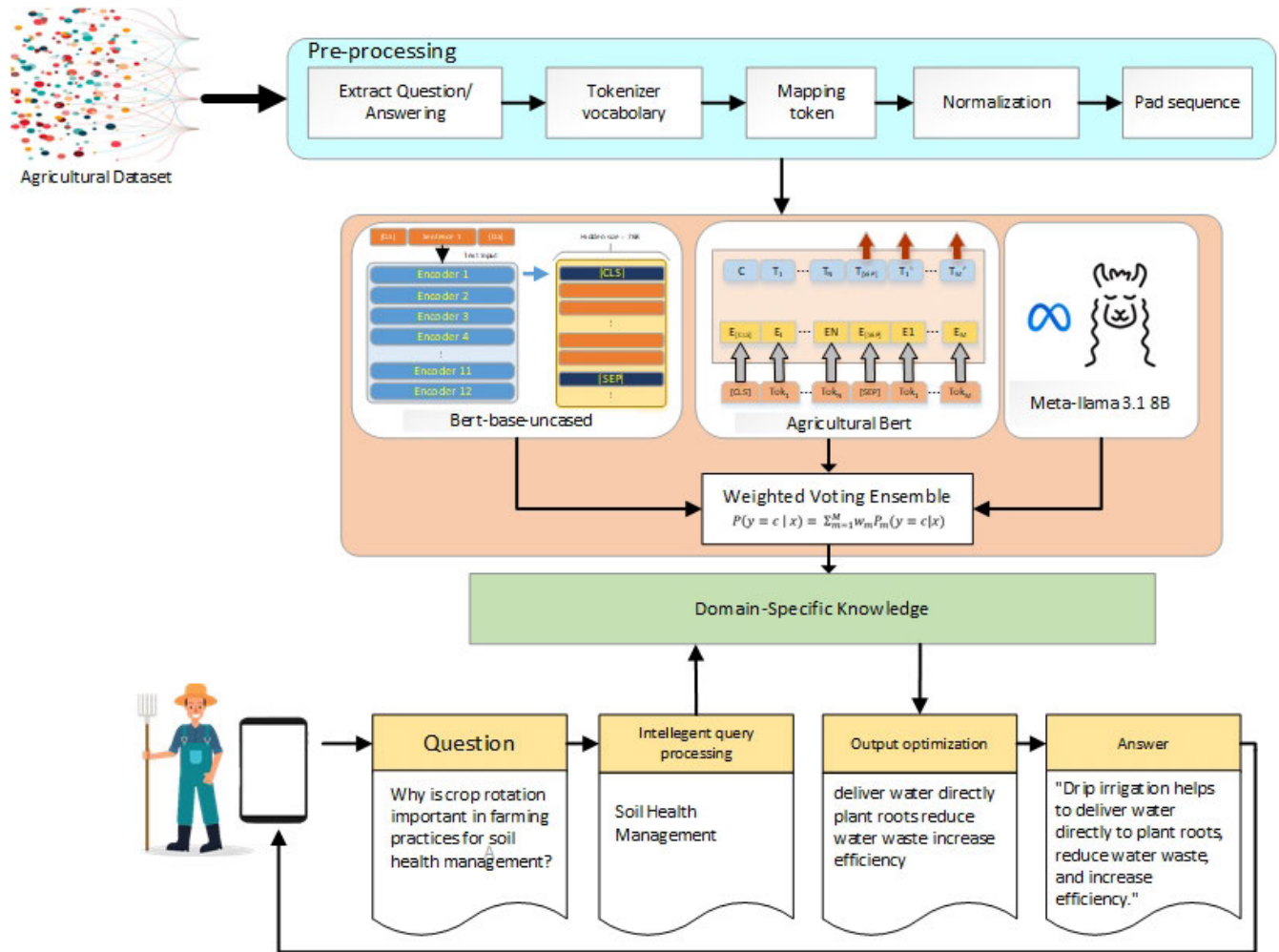


FIGURE 1. Work flow of agricultural knowledge retrieval using large language models for query resolution.

shown to retain a high degree of generality while improving prediction accuracy for a wide range of data in several fields [46].

This paper presents the Chains-BERT model, which further enhances query accuracy for agriculture domain by introducing text-relational chains on top of BERT network to compensate for small corpus sizes and restricted content. Through semi-supervised learning, performance can be improved with a small amount of labeled data, which is suitable for tasks having few annotated instances. In this study, we present an agricultural question knowledge dataset in a dual way: question parsing and the agricultural knowledge base (AgriKB) [47].

III. METHODOLOGY

Our study uses a combination of pre-trained Large Language Models (LLMs) and ensemble techniques. The system is designed to process agricultural queries, extract relevant domain-specific knowledge, and provide optimal answers. The entire process is divided into four key stages:

Pre-processing, ensemble the pre-trained models, intelligent query processing, and Output optimization, shown in figure 1.

A. DATA COLLECTION

The agricultural dataset consists of question-answer pairs, organized in a single-choice format, strategically divided into 80-20 splits for training and testing purposes. Initially containing 22,684 samples, the dataset faced potential overfitting concerns due to its limited size. Additional data was sourced from various agricultural resources, including web content and agricultural forum discussions, to mitigate this risk and enhance robustness. The dataset was further augmented using the BART model from the Hugging Face platform. This approach introduced paraphrased samples to enhance linguistic diversity. By maintaining domain-specific expertise and reducing overfitting risks, the augmentation process significantly improved the dataset's capacity to train models effectively for agricultural knowledge retrieval. This study considers that 30% of the total records are augmented for each category, as reflected in the enhanced dataset's composition.

TABLE 2. Distribution of records across various agricultural categories in the dataset.

Category	Original Records	Augmented Records	Total
Farming	8,390	3,596	11,986
Crop Production	2,761	1,183	3,944
Soil Management	2,202	944	3,146
Pest Management	1,975	847	2,822
Animal Husbandry	323	139	462
Fruit	181	78	259
Grand Total	15,832	6,787	22,685

B. PRE-PROCESSING

The agricultural data, comprising text from various sources, such as web text, soil analysis reports, and crop management guidelines, undergoes several preprocessing steps to make it compatible with LLMs. The pre-processing pipeline includes the following stages. Since 22,684 dataset samples are far from enough for natural language processing problems and can easily lead to a high risk of model overfitting, we collected more real writing data from agricultural website and forums sources. At the same time, we used the source of the huggingface website to generate agriculture data as the final training samples, ensuring the diversity and stability of the model training data.

The agricultural data are processed to generate suitable context in a question-answer format. This step serves as the foundation or prerequisite for training and testing the systems ability to respond effectively to domain-specific queries. As shown in Table 2, this process includes a comprehensive study of the agricultural field and associated concepts such as farming, crop cultivation techniques, soil mechanics or pest management, animal husbandry and fruit.

Within each context, observations are grouped after consulting available agricultural knowledge to determine their relevance to the target areas. If deemed relevant, the system tags and categorizes the information under appropriate themes, such as crop rotation, irrigation methods, or pest control actions. The dataset is filtered using a predefined list of tokens that frequently occur within categories, and questions related to these categories are extracted based on whether they reflect common, practical inquiries that farmers or other agricultural stakeholders are likely to encounter in the field.

We focus on a combination of questions in each domain, from soil health management to advanced irrigation techniques, making sure that our dataset spans a wide array of challenges facing agriculture. In this way, a fully-fledged extraction is constructed to ensure diversity between the questions and answers at all levels, which consequently also increases its domain-specific performance.

The data is converted to structured form to be trained using all the models built earlier. The system takes this processed data and uses it to train a range of different models, each created to understand and answer queries in specific domains. We chose the BERT-based uncased model and

Agriculture-BERT, which are designed for text processes and domain representation. Whereas BERT-based uncased tackles general language understanding, agricultural-BERT aims at agriculture domain-specific wording and context to understand subjects like soil management by speaking about things that sometimes regular people are also confused about.

The Meta-Llama 3.18B model is a state-of-the-art large language model that further improves the complexity and comprehensibility of question generation, understanding, and response for more challenging agricultural questions. This model follows up by analyzing complex and unanswered questions on agriculture practices and impacts of sustainability measure through in-depth contextually resonant answers, shown in Table 3. This combination of models is imposed to utilize ensemble learning using a weighted voting mechanism that takes into account strengths of each unique component. Each model computes a probability that represents how confident it is in the given prediction for any test sample, and during fine-tuning, these probabilities are put together using weights. The prediction with the highest likelihood across all models is multiplied by the said weights to determine the final prediction, ensuring that it returns an accurate and context-appropriate answer.

The weights assigned to each model in the ensemble are selected through empirical analysis and domain expertise. The BERT-based uncased model, which focuses on general language understanding, is assigned a weight of 0.3, as it provide a solid foundation for processing agricultural queries but may lack the domain-specific knowledge required for specific questions. The Agriculture-BERT model, designed specifically for the agricultural domain, is assigned a higher weight of 0.5 due to its ability to capture domain-specific wording and context, which is crucial for providing accurate and relevant answers to agricultural queries. The Meta-Llama 3.18B weighs 0.2. The Agriculture-BERT model, this model's general-purpose character, may lead to lower domain-specific accuracy despite its superior ability to take complex queries and produce detailed solutions. By allocating these weights, the ensemble prioritizes the most pertinent knowledge for the agricultural domain while utilizing each models capabilities.

This helps in utilizing each model to its capacity when solving questions that differ across linguistics and context, assisting the system to be versatile and suited to various problems existing in agricultural practice. The weighted voting ensemble can be expressed mathematically as shown in equation 1:

$$P(Y = c|x) = \sum_{i=1}^n w_i P_i(Y = c|x) \quad (1)$$

n is the number of models in the ensemble. This weighted voting mechanism ensures that models with higher accuracy or domain-specific understanding Bert model have a greater influence on the final prediction.

$$P(y|x, \theta) \quad (2)$$

TABLE 3. The summarization of 3 LLMs running QA models.

Models	Parameters	Description
Bert-based-uncased	110 Millions	This architecture consists of 12 transformer layers or encoder layers, 768 hidden units per layer and 12 attention heads.
AgriBert	110 Millions	The dataset collected is a balance between scientific and available works in the agriculture areas and includes knowledge from other areas of agricultural research and valuable expertise.
Llama3.1	8 Billions	It is designed to provide efficient natural language understanding and generation across various tasks, including question-answering, summarization, and text completion.

represents the probability distribution over possible output given input x and model parameters θ , are the fine-tuned parameters learned by the model during training, shown in equation 2. This final step ensures that the best possible response is provided to the user, offering practical insights into crop management, soil optimization, or pest control.

A knowledge layer associated with the agricultural domain has been integrated into the model to ensure that the responses provided by the system are precise and relevant to the context. Using seasoned farming methods and specialized expertise, this layer functions as a refining and validation process. After extracting question-answer pairs, including domain-specific information is an essential step to ensure that responses meet the practical requirements of farmers and agricultural experts.

C. INTELLIGENT QUERY PROCESSING

To ensure that the system fully comprehends the user's query within the context of agricultural knowledge, this stage employs domain-specific filters and context-aware techniques. Using these instruments, the system can identify important agricultural ideas, including crop rotation, healthy soil, insect control, and irrigation techniques. By dissecting the question into its most basic elements, the system may determine which parts are most pertinent and concentrate on the regions that need to be addressed. The system's overall performance depends on this phase since it guarantees efficient query understanding and answer development. The technology may enable farmers and agricultural experts to make realistic decisions by managing water resources and increasing crop yields, among other complicated agricultural issues, using intelligent query processing.

In the context for my practice of answering the query and producing a response, the system continues to the output optimization phase. This essential step ensures that the completed response is clear, intelligible, and customized to the user's situation. The system considers geographical factors, including modern and local farming methods, especially when answering questions related to crop management. This method guarantees that the answer practically applies to the user's location and is scientifically accurate. Using domain-specific knowledge collected at an earlier stage of the process, the system enhances the response to ensure that it is based on actual farming practices. By improving its relevance and usefulness, this stage ensures that the

information offered satisfies the exact demands of farmers and agricultural experts.

For instance, if a user asks, "What is the best soil for growing cassava?," the system processes the query while integrating soil-related factors specific to cassava cultivation in different regions.

D. OUTPUT OPTIMIZATION

Following the prediction generation, the method proceeds into a crucial output optimization process where the responses produced contrast with pre-established measures in agriculture. It also ensures that the model produces useful and accurate answers in actual agricultural situations. The method validates the accuracy and applicability of its answers by comparing the model outputs with confirmed agricultural practices. The inquiry presents specific agricultural difficulties, which the machine then refines solutions. Farmers will obtain scientifically reliable information that can be directly applied to their conditions thanks to the system's optimization process. This ensures that the final output is succinct, contextually appropriate, and matches the user's practical demands.

Ensemble methods often require training and maintaining multiple models, which can lead to longer training times, especially for large agricultural datasets. Higher computational resource requirements for both training and inference. Increased memory usage to store multiple models. While ensemble methods can reduce overfitting, they can also exacerbate it if not properly implemented. Boosting techniques may focus excessively on difficult examples, potentially leading to overfitting on noise in the agricultural data. Complex ensembles might capture spurious patterns in the training data that do not generalize well to new queries. Careful cross-validation and regularization techniques are crucial to mitigate this risk. In reduced interpretability, As ensemble models combine predictions from multiple base models, it can become challenging to describe how specific agricultural predictions are made. Identify which features or aspects of the query are most influential. Provide clear rationales for recommendations to farmers or agricultural experts. This lack of transparency may reduce trust in the system, especially in critical agricultural decision-making scenarios.

By combining the predictions of multiple models, ensemble learning has become a potent technique for predicting

Algorithm 1 Pseudocode for Training Ensemble Model

Input: Agricultural dataset $X = \{x_1, x_2, \dots, x_n\}$ (Questions) $Y = \{y_1, y_2, \dots, y_n\}$ (Contents)

Output: Train ensemble model M

```

1: Begin
2:  $K$ : Initialize  $K$ : Number of epochs
3: Get data
4: Preprocess the data
5: Set  $n$  to 0
6: Set epoch to 0
7: while epoch <  $K$  do
8:   Set up individual models  $M_2, M_3, WM$ 
9:   while each input data point in  $X$  do
10:    Pass input to models  $M_2, M_3, WM$ 
11:    Calculate predictions  $P_1, P_2, P_3$ 
12:    Set probability weights  $w_1, w_2, w_3$ 
13:    Compute  $WP = (w_1 * P_1 + w_2 * P_2 + w_3 * P_3)$ 
14:   end while
15:   Update weighted voting average ensemble accuracy score
16:   epoch = epoch + 1
17: end while
18: Save Model
19: Load Model
20: while  $n < 3$  do
21:   Set  $M_1, M_2, M_3, WM$  to 0
22:   Pass input to Models
23:   Get probability of  $P_1, P_2, P_3$ 
24:   Set  $w_1, w_2, w_3$ 
25:   Set  $WP = ([w_1, w_2, w_3].[P_1, P_2, P_3])$ 
26:   Get Weighted Voting Ave Ensemble Accuracy_score
27: end while
28: End

```

contents, often outperforming individual models [48]. When capturing content, ensemble approaches generally entail combining the results of several models, like support vector machines, decision trees, and neural networks. Ensemble models can achieve improved accuracy and robustness by utilising numerous classifiers that capture different features of the data and may have varied strengths and limitations. To build these ensemble models, methods including bagging, boosting, and stacking are frequently used. By training each contents on a bootstrapped sample of the training data, bagging produces a variety of models, whereas boosting emphasizes misclassified examples in order to repeatedly improve the performance of weak learners. Through the process of stacking, it gains knowledge base of a higher-level representation of the data by using the predictions of base questions as inputs. Ensemble learning offers a promising approach to advancing state-of-the-art performance in context-based tasks by effectively combining multiple models. The intrinsic parallelizability of the ensemble models allows for effective training and inference on large-scale

image datasets. Consequently, ensemble learning has gained popularity in various NLP application, such as medical and finance analysis.

E. FINETUNING

To provide uniformity across all models, we employed a similar process to one utilized for the LLaMA 3.1 model while fine-tuning the BERT-based-uncased and Agricultural-BERT models. The environments used throughout the fine-tuning procedure were utilized a batch size 16, the learning rate $3e-5$ optimized using learning rate scheduling to minimize overfitting, 20 epochs, and the AdamW optimizer with a weight decay of 0.001. Gradually accumulating gradients were enabled to accommodate the enormous model size and constrained GPU resources.

For the BERT-based-uncased and Agricultural-BERT, the fine-tuning process freezes the initial layers to preserve the pre-trained language understanding while fine-tuning the later layers to focus on agricultural knowledge. Like the LLaMA 3.1 model, supervised fine-tuning was employed, where questions were paired with their corresponding answers from the agricultural dataset. This ensured that the models learned effectively within the structured context of the agricultural domain, improving their performance in generating relevant and precise responses to agricultural queries. Gradient accumulation was enabled to handle the large model size with limited GPU resources. During fine-tuning, the initial layers were frozen to retain pre-trained language capabilities, while the later layers were updated to specialize in agricultural knowledge. This strategy balanced general linguistic understanding with the specialized language and structure of the agricultural dataset, ensuring effective adaptation.

We used a weighted voting ensemble approach to gather predictions from several models, each focused on a distinct facet of agricultural information retrieval. Each model's forecast in a weighted voting ensemble is given a weight according to its performance or reliability, and these weighted predictions are then added together to generate the final result. Using each model's advantages increases accuracy and produces a more thorough and well-rounded response than any one model could. To achieve this, we started a rigorous process of model selection and training. Mathematically, the weighted voting average ensemble can be represented as shown in equation 3:

$$\hat{y}_{\text{ensemble}} = \sum_{i=1}^N w_i \cdot \hat{y}_i \quad (3)$$

where $\hat{y}_{\text{ensemble}}$ represents the ensemble prediction, $\hat{y}_i$ denotes the prediction of the i^{th} base prediction, and w_i signifies the weight assigned to the i^{th} prediction. The weights w_i were established by evaluating each base classifier's performance on a validation set, with the goal of prioritising classifiers that exhibit better individual accuracies.

The weights are carefully adjusted by taking into account the accuracy that each base data achieved on the validation dataset. This results in the best possible combination of predictive strength. The weights of each prediction w_i are calculated by using where Acc_i is the validation accuracy of the i^{th} classifier, in equation 4. Using the combined knowledge from several models, our ensemble method aims to overcome the drawbacks to improve the precision, consistency, and robustness of answer for the users.

$$w_i = \frac{Acc_i}{\sum_{j=1}^N Acc_j} \quad (4)$$

The BLEU score is used in natural language processing (NLP) to compare n-grams between generated and reference text. For this study, we used the sacreBLEU library to calculate BLEU scores by comparing system-generated responses with reference answers provided by agricultural experts. The model's result were specifically evaluated across key agricultural topics, including irrigation, pest management, and crop production, ensuring that the responses were relevant and accurate in domain-specific contexts, as shown on equation 5. We applied the ROUGE metric to provide a more comprehensive evaluation of the quality of the generated texts. ROUGE was used to measure the overlap of n-grams and the longest common subsequence between the model-generated answers and the reference texts. This metric allows for a balanced evaluation of content coverage (recall) and precision, ensuring that the generated responses include relevant content and reflect the structure and substance of the reference answers. Using both BLEU and ROUGE, our objective was to capture a holistic view of the performance of the model, accounting for both accuracy and completeness of the content in the agricultural domain.

$$BLEU = BP \times \exp\left(\sum_{n=1}^N w_n \log P_n\right) \quad (5)$$

$$f(s)_i = \frac{e^{s_i}}{\sum_{j=1}^C e^{s_j}} \quad (6)$$

A vector of scores $s = [s_1, s_2, \dots, s_C]$ is given by the softmax function, which converts it into a probability distribution in which each $f(s)_i$ denotes the probability of the i -th class. It is a valid probability distribution as the sum of all probabilities equals 1, in equation 6.

$$CE = -\sum_{i=1}^C t_i \log(f(s)_i) \quad (7)$$

In this equation 7, the summary covers all types i from 1 to C , and the negative log-likelihood is calculated for the correct class i , weighted by the true label t_i . The objective is to minimize the cross-entropy loss, which means making the predicted probability for the correct class as close to 1 as possible.

TABLE 4. Summary of computing resources used in proposed study.

Computing Resources	Detail
CPU	Intel(R) Core(TM) i7-14700K 3.40 GHz
GPU	NVIDIA GeForce RTX 4080
RAM	700 GB
System	64-bit operating system, x64-based processor
Language	Python(3.9.19)
Framework	Pytorch 11.2 backend
Software	Anaconda

IV. RESULTS

A. ENVIRONMENT SETUP

In order to guarantee accurate and effective research, we set up a dedicated workspace for this purpose. In Table 4 shown provides a comprehensive overview of our environment configuration, including all individual settings. This method improved the validity and dependability of our research findings by allowing us to perform a complete investigation and analysis.

B. EVALUATION METRICS

We used accuracy, precision, recall and f1-score to identify the performance of our model performance. The predicted results are denoted by the data.

1) ACCURACY

Accuracy measure of the model is defined by dividing the total number of instances analyzed by the proportion of adequately predicted cases, both true positives and true negatives, the equation in 8.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

2) F1 SCORE

F1 scores are useful in cases of data imbalance. Recall and accuracy are harmonically mean, and F1 scores offer a single metric to balance them both, in equation 9.

$$\text{F1 Score} = \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

3) RECALL

Sensitivity or recall is the fraction of true predicted values to actual positives, the equation in 10.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

4) PRECISION

Precision tells how accurate the model made the positive predictions. It is the fraction that correctly predicts true positive and total predictive positives, in equation 11.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

Our ensemble model technique integrates predictions from several models to improve performance. The high accuracy of the model and the F1 scores demonstrate

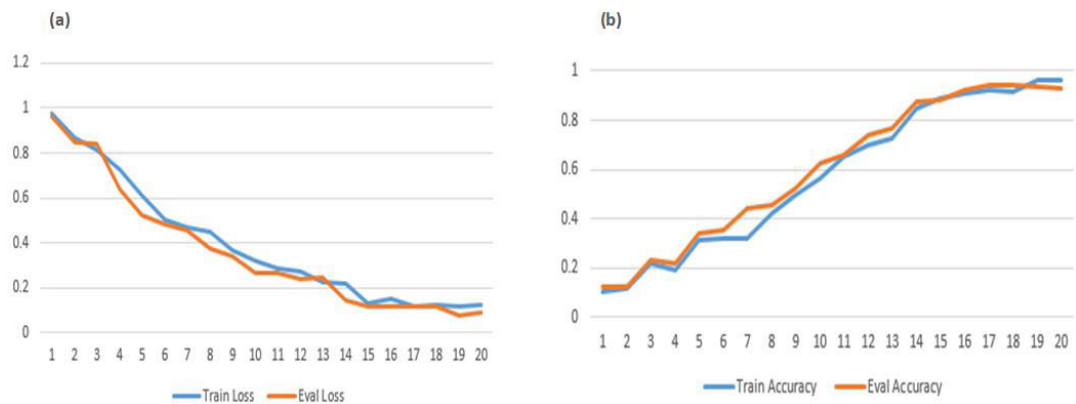


FIGURE 2. (a) Training and evaluation loss consistently decreases, indicating effective learning. (b) Training and evaluation accuracy steadily improve, reflecting enhanced model performance.

TABLE 5. Effectiveness comparison of various models, highlighting the enhanced performance of the ensemble model over individual models.

Models	Accuracy	Recall	Precision	F1-Score
Bert-based-uncased	84.09%	78.62%	81.41%	80.02%
AgriBert	86.44%	81.71%	82.62%	82.16%
RoBERTa	88.34%	83.59%	85.90%	84.72%
DistilBERT	83.71%	79.92%	82.48%	81.18%
Longformer	86.28%	81.67%	84.75%	83.16%
CamemBERT	85.16%	81.43%	82.23%	81.82%
Llama3.1	91.56%	85.62%	88.86%	87.16%
Our Ensemble Model	93.10%	83.61%	86.57%	85.04%

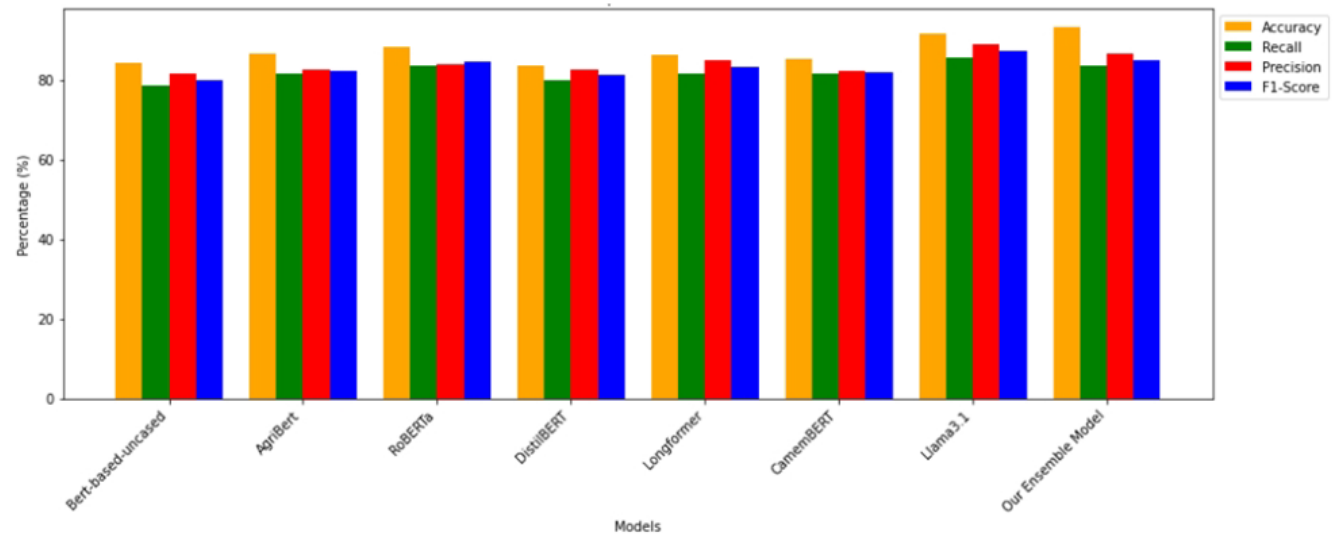


FIGURE 3. Comparison of models accuracy and F1-score.

its substantial capacity to handle agricultural inquiries, demonstrating that each model significantly contributes to comprehending and reacting to domain-specific data. However, combining knowledge from several sources, lowering errors and improving answer relevance, the ensemble model's higher performance highlights its capacity to provide more dependable and contextually appropriate responses.

When evaluating performance in knowledge retrieval tasks, metrics such as recall and accuracy are crucial, especially when large language models are utilized for question-answering or data extraction. While precision evaluates the correctness of the information obtained, ensuring that responses are relevant and accurate, recall measures the proportion of relevant information that the LLM successfully

TABLE 6. Summary our approach using in Bleu and Rouge score.

Model	Bleu Score	ROUGE-1	ROUGE-2	ROUGE-L
BERT-base-uncased	45.5	0.62	0.45	0.53
Agricultural-BERT	48.2	0.65	0.48	0.56
LlaMA 3.1 (8B)	50.3	0.68	0.51	0.58
Ensemble Model	53.8	0.70	0.55	0.61

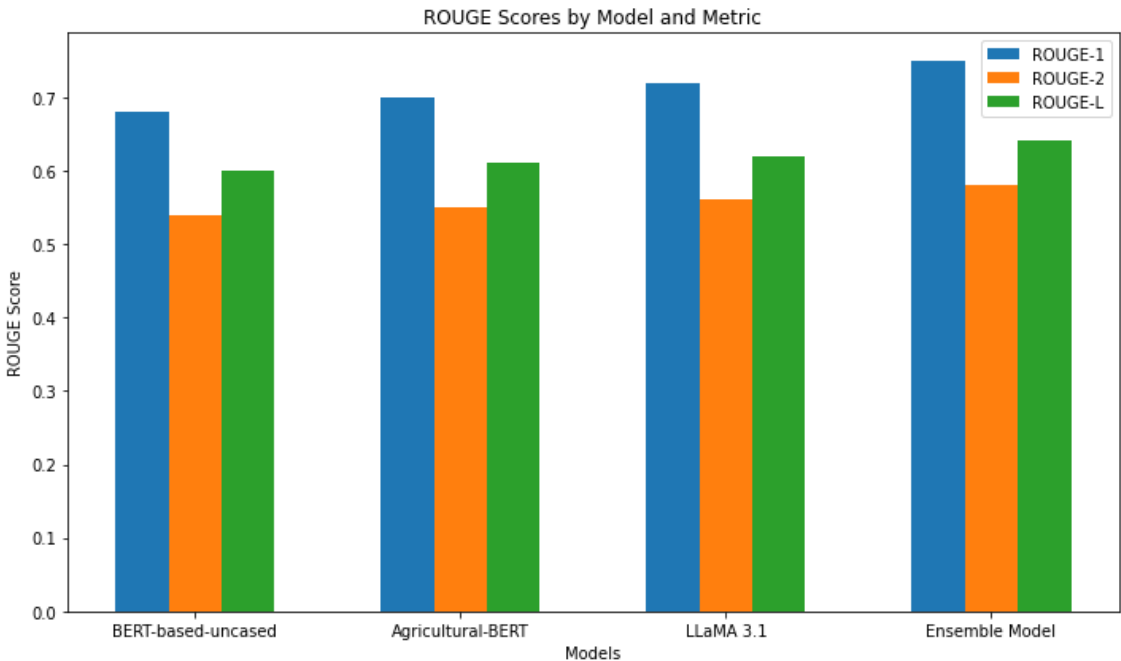


FIGURE 4. Comparison of ROUGE scores across different models.

recovers, demonstrating its ability to properly identify essential elements. These indicators confirm that the ensemble model is a valuable tool for facilitating decision making and information exchange in the field by providing high quality, domain-specific agricultural data.

As shown in Figure 2 (a), it demonstrates consistent convergence across 20 epochs. The steady decline in both curves indicates effective error reduction and progression of learning. The close alignment between training and evaluation loss suggests good generalization with minimal overfitting, as the model performs similarly on both seen and unseen data. In Figure 2 (b) show systematic improvement in both training and evaluation metrics. The gradual ascent and eventual convergence near optimal accuracy indicates stable learning dynamics. The tight correlation between training and evaluation accuracy further validates the robust generalization capabilities of the model.

As shown in table 5, presents a performance comparison of different language models on key classification metrics: accuracy, recall, precision, and F1-score. Among the individual models, Llama3.1 achieves the highest accuracy (91.56%) and F1-score (87.16%), indicating its strength in overall prediction quality and its balance between precision and recall. Longformer and RoBERTa also perform well, with

TABLE 7. Weight distribution comparison.

Weight Distribution Bert-Uncased/Agricultural Bert/ Meta Llama 3.1	Accuracy	F1-Score
0.3/0.5/0.2	93.10%	85.04%
0.4/0.4/0.2	81.23%	73.16%
0.2/0.6/0.2	80.56%	72.48%

F1-scores of 83.16% and 84.72%, respectively, reflecting their reliability in classification tasks. DistilBERT and Bert-based-uncased show relatively lower performance, with F1-scores of 81.18% and 80.02%, respectively, suggesting that they may be less effective for this task compared to other models. The ensemble model, which combines multiple models, outperforms all individual models with an accuracy of 93. 10% and an F1 score of 85. 04%, demonstrating that an ensemble approach can enhance predictive performance by taking advantage of the strengths of multiple models. This suggests that the ensemble model is the most robust choice, achieving the best balance across all metrics evaluated.

The performance of the ensemble model was tested using different weight configurations, as shown in Table 7. The configuration with weights of 0.3/0.5/0.2 for BERT-based-uncased, Agricultural-BERT and LLaMA 3.1, respectively,

TABLE 8. Comparison of various language models, highlighting their specialized use cases and general challenges.

Reference	Year	Model	Accuracy	Advantage	Limitation
[49]	2024	AgroGPT	89.41%	Utilizing expert-tuning and specialized datasets, this agriculture-focused study outperforms generic models in accuracy, particularly for fine-grained tasks like disease and insect detection.	Lack of fine-grained datasets for concepts like insects and diseases, and the need for further improvement in real-world field applications.
[50]	2024	KITA	91.5%	Enhances controllability and dependability of task-oriented conversation mechanisms by fusing programmable rules with large language models.	The samples given during fine-tuning impact performance, and if the data is inadequate or unrepresentative, they might not generalize well to all situations.
[51]	2024	Bert Based	80.67%	A refined retrieval augmentation method allows large language models to answer document questions with more significant domain flexibility, accuracy, and reliability, decreasing hallucinations and enhancing their ability to provide relevant, well-founded responses.	Restrictions like reliance on pre-trained models, limited domain generalization, manual dataset creation, possible computational overhead, limited benchmarking, and domain-specific overfitting risk.
[52]	2023	Flan-T5	78.5%	This LLM method offers reduced complexity, near-state-of-the-art implementation, computational efficiency, scalability, and excellent generalizability, making it entirely effective for text-dense document question-answering tasks.	Its inability to process layout-based data, reliance on correct reading order, difficulty with lengthy texts, and limitation to responding only with extracted text, losing essential visual context when needed.
[53]	2024	GPT4	80%	This framework for subpopulation findings. Addressing biases improves dataset interpretability and model robustness, making it valuable for equitable, data-centric improvements across applications like slice discovery and subpopulation shift.	Increased computing needs, potential model-induced biases, limited human validation, and dependence on the accuracy and biases of underlying models are some of the model's shortcomings that could limit its relevance and nuanced accuracy in various settings.

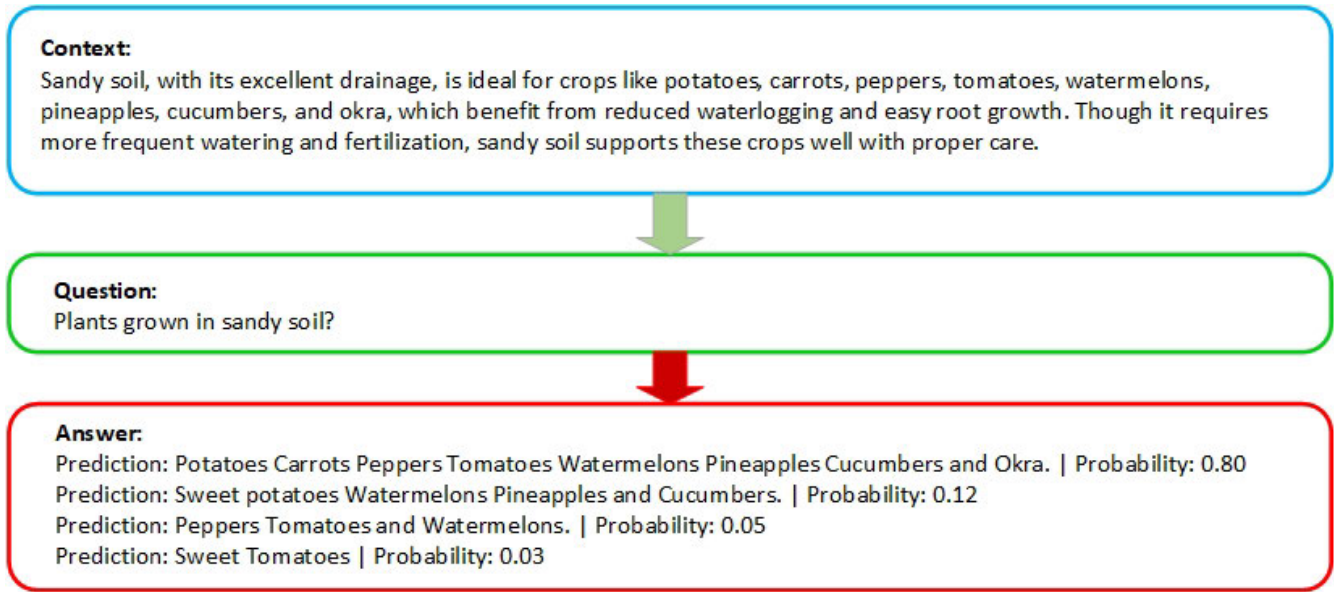


FIGURE 5. Context-based predictions for crops suited to sandy soil, with probability scores assigned to each prediction, highlighting the first prediction as the most accurate response with the highest confidence score.

yielded the highest accuracy of 93.10% and F1-score of 85.04%. These findings validate the chosen weight distribution, which leverages the Agricultural-BERT domain-specific accuracy while maintaining contributions from the other models. Our ensemble model demonstrated its ability to generate precise, domain-specific, contextually relevant responses that closely resemble reference texts by achieving a BLEU score of 53.8 and strong ROUGE scores (ROUGE-1: 0.70, ROUGE-2: 0.55, and ROUGE-L: 0.61). The model is well-suited for agricultural knowledge retrieval, as seen

by the BLEU score, which emphasizes its accuracy in replicating key terms and phrases, while the ROUGE scores highlight its coverage of important information. Using the advantages of each unique model, the ensemble offers comprehensive and pertinent responses to intricate and subject-specific agricultural questions. This enhancement, as shown in Table 6, demonstrates the ensemble model's ability to produce improved results that satisfy the real-world requirements of agricultural experts. The ensemble approach improves accuracy and strength by utilizing the advantages

of individual models. The method results in a more thorough and well-rounded response by including various viewpoints from each component model. This ability to deliver comprehensive, pertinent responses is beneficial for intricate, subject-specific agricultural questions where accuracy and comprehensiveness of data are essential.

Shown in figure 5, In the prompt “Plants grown in sandy soil?,” our model made several predictions. The probability score assigned to each prediction indicated the model’s confidence level in its response. With a probability of 0.80, “Potatoes Carrots Peppers Tomatoes Watermelons Pineapples Cucumbers and Okra” was the most likely prediction, showing high model confidence in this response. Predictions such as “Sweet potatoes Watermelons, Pineapples and Cucumbers” with a probability of 0.12 and “Peppers Tomatoes and Watermelons” with a probability of 0.05 had far lower probabilities, suggesting less confidence. “Sweet Tomatoes” was the last prediction, with the lowest probability of 0.003, indicating that the model thought it was the least likely choice. These probabilities shed light on the distribution of possible responses to agricultural questions and help in quantifying the model’s uncertainty.

V. DISCUSSION

The results of this analysis show the value of our suggested weighted voting ensemble method for enhancing large language model (LLM), based agricultural information retrieval. In terms of accuracy and contextual relevance, the ensemble model significantly outperformed individual models by using the capabilities of numerous high-performing pre-trained models. With an accuracy of 93.1%, the ensemble model outperformed other methods such as other LLM models. The ensemble method’s weighted voting process, which gives more weight to models that have improved individual performance and produce a final forecast that is more accurate and well-rounded, is responsible for this improvement. As shown in Table 8, although other models, including AgroGPT and KITA, perform well in particular domains, the ensemble model’s capacity to combine these many advantages improves precision and applicability for agricultural inquiries.

These increased BLEU and ROUGE scores demonstrate how well the model delivers correct and contextually appropriate agricultural responses by providing language that nearly resembles reference responses. In contrast to ROUGE, which emphasizes the model’s coverage of crucial information and shows its fitness for agricultural knowledge retrieval, BLEU evaluates the degree of n-gram overlap and emphasizes accuracy in recreating essential terms and phrases from reference texts.

Moreover, the model currently works in English, which limits its use in areas with several languages. It would be more globally applicable if the system managed datasets in several languages, especially in regions where agricultural operations occur in linguistically and culturally diverse environments.

Adding multilingual support could allow the model to reach a wider audience and provide more accessible and understandable information to farmers and other agricultural experts. Further investigation may include sophisticated ensemble methods such as stacking or boosting to improve the accuracy and resilience of the model. Enhancing the model’s capacity to represent intricate agricultural information may help it perform better across various activities and domains. Putting these improvements into practice would increase the model’s adaptability for use in the global agricultural domain.

VI. CONCLUSION AND FUTURE WORK

In this study, we proposed a new way to improve agricultural knowledge retrieval using large language models (LLM) and ensemble learning. This system is intended to answer crop management, disease detection, and other agricultural queries accurately. We fine-tuned our ensemble models on the domain-specific data and achieved substantial improvement in query resolution, with the highest accuracy of 93% results utilize weighted voting. The findings suggest that combining domain-specific data with LLM can provide insightful information to farmers, enabling them to make more evidence-based decisions and consequently increase agricultural output. The proposed system detects agricultural diseases and provides useful information for crop production questions, enabling farmers to make informed decisions to get better yields and manage resources properly.

Our analysis of our study using large language model ensembles shows several limitations, including dataset constraints, limited real-world testing, and high computational costs, indicating opportunities for future research. Implementing multimodal data, model distillation, multilingual capabilities, and explainable AI will enhance the accuracy, accessibility, and usability of the system. They will significantly improve agricultural decision making in various global contexts.

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CYRENEO DOFITAS JR received the M.S. degree in computer science from Iloilo Science and Technology University, Iloilo City, Philippines. Currently, he is pursuing the Ph.D. degree in electronic engineering with Jeju National University, South Korea. He is a Research Fellow with the Machine Learning Laboratory, Jeju National University. His research interests include machine learning, video processing, deep learning, and large language models.



YONG-WOON KIM was born in Seoul, South Korea, in 1971. He received the M.S. degree in computer science from Yonsei University, South Korea, in 1997, and the Ph.D. degree in computer science and engineering from Christ University, Bengaluru, India. He was an Associate Professor and a Coordinator with the Centre for Digital Innovation, Christ University. He is currently a Visiting Professor with Jeju National University, South Korea. Before joining Jeju

National University, he was with Samsung Electronics Company Ltd., LG Electronics Company Ltd., and many global IT companies as a Principal Embedded Software Engineer for ten years and has been an Entrepreneur in India, since 2007. He completed research collaborations with the Electronics and Telecommunications Research Institute (ETRI) and other global research and development organizations. His research interests include computer vision, artificial intelligence, and the Internet of Things.



YUNG-CHEOL BYUN received the B.S. degree from Jeju National University in 1993 and the M.S. and Ph.D. degrees from Yonsei University, in 1995 and 2001, respectively. He was a Special Lecturer with SAMSUNG Electronics and SDS, from 1998 to 2001. From 2001 to 2003, he was a Senior Researcher with the Electronics and Telecommunications Research Institute (ETRI). He was promoted to join Jeju National University as an Assistant Professor, in 2003. Currently,

he is a Full Professor with the Computer Engineering Department, Jeju National University. His research interests include the areas of AI & machine learning, pattern recognition, blockchain and deep learning-based applications, big data and knowledge discovery, time-series data analysis and prediction, image processing, medical image applications, and recommendation systems.

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